Lecture 8: Gelfand Triples and Abstract Parabolic Equations

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04.12.2025

Chapter 3: Variational approach to SPDEs (continued)

Gelfand triples

Let V and H be separable Hilbert spaces with a dense and continuous embedding

$$V \hookrightarrow H.$$

Concretely, "continuous embedding" means that there is a constant $C > 0$ such that

$$\|u\|_H \leq C\|u\|_V, \qquad u \in V,$$

so the V-norm is stronger than the H-norm (more regularity in V implies control in H). Without loss of generality, we may assume that

$$\|u\|_H \leq \|u\|_V, \qquad u \in V.$$

This normalization is harmless: if originally $\|u\|_H \leq C\|u\|_V$, one can replace the norm on V by the equivalent norm $\|u\|_{V,\text{new}} := C\|u\|_V$. Equivalently, one may extend the V-norm to all of H by setting $\|u\|_V := \infty$ for $u \in H \setminus V$; then the inequality above holds for every $u \in H$. By identifying H with its dual H' via the Riesz isomorphism, we obtain the *Gelfand triple*

$$V \hookrightarrow H \cong H' \hookrightarrow V'.$$

The meaning of the last embedding $H' \hookrightarrow V'$ is that every functional on H restricts to a functional on V; this restriction is continuous because $V \hookrightarrow H$ is continuous. We write ${}_{V'}\langle \cdot, \cdot \rangle_V$ for the duality pairing between V' and V; it extends the inner product on H in the sense that

$${}_{V'}\langle h, v \rangle_V = \langle h, v \rangle_H, \qquad h \in H,\ v \in V.$$

Remark 1 (Typical example). For a bounded domain $D \subset \mathbb{R}^d$,

$$V = H_0^1(D), \qquad H = L^2(D), \qquad V' = H^{-1}(D).$$

In this example, V is the energy space, H is the pivot space with the L^2 inner product, and V' is the corresponding dual space; concretely, $H_0^1(D)$ is the Sobolev space of L^2-functions with weak first derivatives in L^2 and zero boundary values, while $H^{-1}(D)$ is its dual, the negative Sobolev space.

Remark 2 (Why one says "space regularity"). In PDE applications, the space V usually describes regularity in the spatial variable. For instance, if $V = H_0^1(D)$, then $u(t) \in V$ means that for each fixed time t the function $x \mapsto u(t, x)$ has one weak spatial derivative in $L^2(D)$, so membership in $L^2([0, T]; V)$ expresses spatial regularity integrated over time.

Operators $A : V \to V'$

Let $A : V \to V'$ be a (typically linear) operator. Common assumptions in the variational theory are:

- **Boundedness:** $A \in \mathcal{L}(V, V')$, i.e. there exists $C > 0$ such that

  $$\|Au\|_{V'} \leq C\|u\|_V, \qquad u \in V.$$

  A lot of the time, the operators arising in PDEs are unbounded when viewed as operators on H. In the variational framework, the relevant condition is that A is bounded as a map from V into V'. Intuitively, boundedness means that A does not amplify the size of u in an uncontrolled way: if u is small in V, then Au is small in V'.

- **Coercivity (bounded from below):** there exist $\alpha > 0$ and $\lambda \in \mathbb{R}$ such that

  $${}_{V'}\langle Au, u\rangle_V \geq \alpha\|u\|_V^2 - \lambda\|u\|_H^2, \qquad u \in V.$$

  Intuitively, coercivity says that A controls the stronger V-norm of u up to a lower-order correction involving the weaker H-norm; this is the key estimate behind energy bounds. Depending on the sign convention in the evolution equation, one often also encounters the equivalent "dissipative" form

  $${}_{V'}\langle Au, u\rangle_V \leq -\alpha\|u\|_V^2 + \lambda\|u\|_H^2,$$

  which is obtained by replacing A with $-A$ (e.g. writing the heat equation as $u_t - \Delta u = f$ rather than $u_t = \Delta u + f$).

- **Monotonicity:** for all $u, v \in V$,

  $${}_{V'}\langle Au - Av, u - v\rangle_V \geq 0.$$

  Intuitively, monotonicity means that A does not push two states farther apart in a harmful direction; it is a nonlinear analogue of positivity and is central for uniqueness and stability.

Example: the Laplacian

Take $V = H_0^1(D)$ and define $A : V \to V'$ by

$${}_{V'}\langle Au, v\rangle_V := \int_D \nabla u \cdot \nabla v \, dx, \qquad u, v \in H_0^1(D).$$

Formally, this corresponds to $Au = -\Delta u$ with homogeneous Dirichlet boundary conditions, since for smooth u, v,

$$\int_D \nabla u \cdot \nabla v \, dx = -\int_D (\Delta u)\, v \, dx.$$

Moreover, this definition shows directly that A is bounded from V to V': for $u \in V$,

$$\|Au\|_{V'} = \sup_{\|v\|_V \leq 1} {}_{V'}\langle Au, v\rangle_V = \sup_{\|v\|_V \leq 1} \int_D \nabla u \cdot \nabla v \, dx \leq \sup_{\|v\|_V \leq 1} \|\nabla u\|_{L^2(D)} \|\nabla v\|_{L^2(D)} \leq \|u\|_V.$$

Here we used Cauchy–Schwarz and the fact that $\|v\|_V$ controls $\|\nabla v\|_{L^2(D)}$ for $v \in H_0^1(D)$.

Abstract parabolic Cauchy problem

Theorem 1 (Existence and uniqueness in a Gelfand triple)**.** Let $A \in \mathcal{L}(V, V')$ be coercive, let $u_0 \in H$, and let $f \in L^2([0,T]; V')$. Then the evolution problem

$$\begin{cases} u'(t) = Au(t) + f(t), & t \in (0,T), \\ u(0) = u_0, \end{cases}$$

has a unique solution

$$u \in C([0,T]; H) \cap L^2([0,T]; V), \qquad u' \in L^2([0,T]; V').$$

In words, u has space-time regularity: it is continuous in time with values in H and square-integrable in time with values in the more regular space V, while its time derivative is square-integrable with values in the dual space V'.

Remark 3 (Heat equation)*.* With $V = H_0^1(D)$, $H = L^2(D)$, and $A = -\Delta$, the abstract problem reads

$$u'(t) = \Delta u(t) + f(t), \qquad u(0) = u_0,$$

and the solution satisfies $u(t) \in H_0^1(D)$ for a.e. t and $u \in C([0,T]; L^2(D))$. This means that for almost every fixed time t, the spatial function $x \mapsto u(t,x)$ belongs to $H_0^1(D)$, i.e. it has weak first derivatives in $L^2(D)$ and zero boundary values.

Remark 4 (Why this is called parabolic)*.* If $A = -\Delta$, then the abstract equation is the heat equation

$$u_t - \Delta u = f.$$

In one space dimension this becomes $u_t - u_{xx} = f$, whose second-order part has discriminant $A = -1$, $B = 0$, $C = 0$, hence $B^2 - AC = 0$, so it is parabolic in the classical classification. In higher dimensions one says it is parabolic because it is first order in time and its spatial second-order part $-\Delta$ is elliptic.

Uniqueness: energy estimate

To prove uniqueness, let u and v be two solutions in the class

$$C([0,T]; H) \cap L^2([0,T]; V), \qquad u', v' \in L^2([0,T]; V'),$$

with the same initial condition $u(0) = v(0)$. Set $w := u - v$. Then w solves the corresponding homogeneous parabolic equation

$$\begin{cases} w'(t) = Aw(t), & t \in (0,T), \\ w(0) = 0. \end{cases}$$

The key identity is the following.

Lemma 1 (Derivative of the H-norm)**.** If $w \in L^2([0,T]; V)$ and $w' \in L^2([0,T]; V')$, then the map $t \mapsto \|w(t)\|_H^2$ is absolutely continuous and for a.e. $t \in (0,T)$,

$$\frac{d}{dt}\|w(t)\|_H^2 = 2\,{}_{V'}\langle w'(t), w(t)\rangle_V.$$

Applying this with $w' = Aw$ yields, for a.e. t,

$$\frac{d}{dt}\|w(t)\|_H^2 = 2\,{}_{V'}\langle Aw(t), w(t)\rangle_V.$$

If, for example, A satisfies the dissipative estimate

$${}_{V'}\langle Aw(t), w(t)\rangle_V \le \lambda \|w(t)\|_H^2,$$

Here $\lambda \in \mathbb{R}$ is a constant measuring how much growth is still allowed in the H-energy. Intuitively, this one-sided estimate says that A does not increase the size of w too fast in H. This is different from coercivity, which gives a lower bound on ${}_{V'}\langle Au, u\rangle_V$ and controls the stronger V-norm of u. then

$$\frac{d}{dt}\|w(t)\|_H^2 \le 2\lambda \|w(t)\|_H^2.$$

Equivalently,

$$\frac{d}{dt}\Big(e^{-2\lambda t}\|w(t)\|_H^2\Big) \le 0.$$

Hence the map $t \mapsto e^{-2\lambda t}\|w(t)\|_H^2$ is nonincreasing. Since $w(0) = 0$, we obtain

$$\|w(t)\|_H^2 = 0 \qquad \text{for all } t \in [0, T].$$

Hence $w \equiv 0$, and therefore $u = v$.

Existence via approximation (Galerkin method)

Let $(e_k)_{k\ge1}$ be a complete orthonormal system (CONS: C means $\overline{\operatorname{span}\{e_1, e_2, \dots\}} = H$, O means $\langle e_i, e_j\rangle_H = \delta_{ij}$, and S means that $(e_k)_{k\ge1}$ is a family of elements of H) of H such that $e_k \in V$ for all k. For $n \in \mathbb{N}$ define the finite-dimensional subspace

$$V_n := \operatorname{span}\{e_1, \dots, e_n\} \subset V.$$

We project the equation onto V_n and look for an approximate solution $u_n \in C([0, T]; V_n)$ of the form

$$u_n(t) = \sum_{k=1}^{n} g_k^{(n)}(t)\, e_k,$$

where the scalar functions $g_k^{(n)}$ are the *Fourier coefficients* of u_n in the basis (e_k). Then u_n is the unique solution in $C([0, T]; V_n)$ of an n-dimensional linear ordinary differential equation. More precisely, testing the evolution equation against e_k, $k = 1, \dots, n$, yields the finite-dimensional system

$$\begin{cases} \dfrac{d}{dt}\langle u_n(t), e_k\rangle_H = {}_{V'}\langle Au_n(t), e_k\rangle_V + {}_{V'}\langle f(t), e_k\rangle_V, & t \in (0, T), \\ \langle u_n(0), e_k\rangle_H = \langle u_0, e_k\rangle_H, \end{cases} \qquad k = 1, \dots, n.$$

This is an n-dimensional Cauchy problem for the coefficient vector

$$\big(g_1^{(n)}, \dots, g_n^{(n)}\big),$$

so by the standard existence and uniqueness theory for linear ODEs it has a unique solution.

Remark 5 (On choosing the basis (e_k)). Since V is dense in H, one can choose a CONS (e_k) with $e_k \in V$ (e.g. by applying Gram–Schmidt to a dense sequence in V). In PDE examples it is common to take (e_k) as eigenfunctions of an elliptic operator (such as the Dirichlet Laplacian), which makes the Galerkin system coincide with a truncated Fourier expansion.

Remark 6 (What elliptic means here). For a second-order linear operator

$$Lu = -\sum_{i,j=1}^{d} a_{ij}(x)\,\partial_{ij}u + \text{lower-order terms},$$

elliptic means that the coefficient matrix $A(x) = (a_{ij}(x))$ is positive definite, i.e.

$$\sum_{i,j=1}^{d} a_{ij}(x)\xi_i\xi_j > 0 \qquad \text{for every } \xi \in \mathbb{R}^d \setminus \{0\}.$$

For the Laplacian, $a_{ij}(x) = \delta_{ij}$, so $-\Delta$ is elliptic.

Uniform energy estimate (a priori bound)

It is called an *a priori* estimate because the bound is derived from the equation and the data alone, before knowing the solution explicitly.

The next step is to pass to the limit $n \to \infty$. The essential ingredient for this is an a priori estimate that is uniform in n. Testing the Galerkin equation with $u_n(t) \in V_n$ and using coercivity yields an energy inequality of the form

$$\sup_{n\geq 1}\left(\sup_{t\in[0,T]} \|u_n(t)\|_H^2 \;+\; \int_0^T \|u_n(t)\|_V^2\,dt\right) < \infty,$$

In words, all approximate solutions u_n satisfy the same bound: their H-norms stay bounded for all times $t \in [0,T]$, and their V-norms are square-integrable in time. The constant in this estimate depends only on $\|u_0\|_H$, $\|f\|_{L^2([0,T];V')}$, and the constants in the assumptions on A, but not on n.

Remark 7 (Heat equation: why Sobolev spaces appear). For the heat equation on a domain $D \subset \mathbb{R}^d$ with $V = H_0^1(D)$ and $H = L^2(D)$, the V-norm is (up to equivalence) the L^2-norm of the gradient:

$$\|u\|_V^2 \simeq \|\nabla u\|_{L^2(D)}^2.$$

Here $\simeq$ means that the two quantities are comparable up to multiplicative constants: each one bounds the other. This follows from Poincaré's inequality on $H_0^1(D)$.

Remark 8 (Why this follows from Poincaré's inequality). If we take

$$\|u\|_V^2 = \|u\|_{H_0^1(D)}^2 := \|u\|_{L^2(D)}^2 + \|\nabla u\|_{L^2(D)}^2,$$

then Poincaré's inequality gives

$$\|u\|_{L^2(D)} \leq C_P\|\nabla u\|_{L^2(D)}, \qquad u \in H_0^1(D).$$

Therefore

$$\|u\|_V^2 = \|u\|_{L^2(D)}^2 + \|\nabla u\|_{L^2(D)}^2 \leq (C_P^2 + 1)\|\nabla u\|_{L^2(D)}^2.$$

Conversely,

$$\|\nabla u\|_{L^2(D)}^2 \le \|u\|_V^2.$$

This is immediate from the definition of $\|\cdot\|_V$, since $\|u\|_V^2$ is the sum of $\|u\|_{L^2(D)}^2$ and $\|\nabla u\|_{L^2(D)}^2$. Hence

$$\|\nabla u\|_{L^2(D)}^2 \le \|u\|_V^2 \le (C_P^2+1)\|\nabla u\|_{L^2(D)}^2,$$

which is exactly the meaning of

$$\|u\|_V^2 \simeq \|\nabla u\|_{L^2(D)}^2.$$

For the heat equation, this gives a concrete example of the uniform a priori estimate: for the Galerkin sequence (u_n),

$$\sup_{n\ge 1}\left(\sup_{t\in[0,T]}\|u_n(t)\|_{L^2(D)}^2 + \int_0^T \|\nabla u_n(t)\|_{L^2(D)}^2\,dt\right) < \infty.$$

This is why one works in Sobolev spaces: for $u \in H_0^1(D)$, the gradient ∇u is defined in the weak sense, and the estimate gives control of these weak derivatives.

Energy estimate: derivation

We now prove the uniform a priori estimate for the Galerkin approximations (u_n).

Starting from the Galerkin system, one computes (formally; justified since u_n is finite-dimensional and smooth in time) Since $(e_1,\dots,e_n)$ is an orthonormal basis of V_n and $u_n(t) \in V_n$, Parseval's identity gives

$$\|u_n(t)\|_H^2 = \sum_{k=1}^n \langle u_n(t), e_k\rangle_H^2.$$

Therefore

$$\frac{1}{2}\frac{d}{dt}\|u_n(t)\|_H^2 = \frac{1}{2}\frac{d}{dt}\sum_{k=1}^n \langle u_n(t), e_k\rangle_H^2 = \sum_{k=1}^n \langle u_n'(t), e_k\rangle_H\,\langle u_n(t), e_k\rangle_H.$$

Using $u_n'(t) = Au_n(t) + f(t)$ and the defining property of the Galerkin approximation,

$$\sum_{k=1}^n \langle u_n'(t), e_k\rangle_H\,\langle u_n(t), e_k\rangle_H = \sum_{k=1}^n {}_{V'}\langle Au_n(t) + f(t), e_k\rangle_V\,\langle u_n(t), e_k\rangle_H$$

Since

$$u_n(t) = \sum_{k=1}^n \langle u_n(t), e_k\rangle_H\, e_k,$$

linearity of the duality pairing in the second argument gives

$$\sum_{k=1}^n {}_{V'}\langle Au_n(t)+f(t), e_k\rangle_V\,\langle u_n(t), e_k\rangle_H = {}_{V'}\langle Au_n(t)+f(t), \sum_{k=1}^n \langle u_n(t), e_k\rangle_H\, e_k\rangle_V = {}_{V'}\langle Au_n(t)+f(t), u_n(t)\rangle_V.$$

Hence

$$\frac{1}{2}\frac{d}{dt}\|u_n(t)\|_H^2 = {}_{V'}\langle Au_n(t), u_n(t)\rangle_V \;+\; {}_{V'}\langle f(t), u_n(t)\rangle_V.$$

Assuming the simplified coercivity ${}_{V'}\langle Au, u\rangle_V \le -\alpha\|u\|_V^2$ (or after rewriting A by shifting a lower-order term), and applying Cauchy–Schwarz in (V', V) followed by Young's inequality,

$${}_{V'}\langle f(t), u_n(t)\rangle_V \le \|f(t)\|_{V'}\|u_n(t)\|_V \le \frac{1}{2\alpha}\|f(t)\|_{V'}^2 + \frac{\alpha}{2}\|u_n(t)\|_V^2,$$

where we used Young's inequality in the general form

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}, \qquad a, b \ge 0, \quad \frac{1}{p} + \frac{1}{q} = 1,$$

or, in the weighted quadratic case relevant here,

$$ab \le \frac{1}{2\varepsilon} a^2 + \frac{\varepsilon}{2} b^2.$$

Here we take $p = q = 2$, $a = \|f(t)\|_{V'}$, $b = \|u_n(t)\|_V$, and $\varepsilon = \alpha$; equivalently, one may view this as the weighted AM–GM inequality. Therefore

$$\frac{1}{2}\frac{d}{dt}\|u_n(t)\|_H^2 \le -\alpha\|u_n(t)\|_V^2 \; + \; \frac{1}{2\alpha}\|f(t)\|_{V'}^2 \; + \frac{\alpha}{2}\|u_n(t)\|_V^2 = -\frac{\alpha}{2}\|u_n(t)\|_V^2 \; + \; \frac{1}{2\alpha}\|f(t)\|_{V'}^2.$$

Integrating from 0 to t gives

$$\frac{1}{2}\|u_n(t)\|_H^2 - \frac{1}{2}\|u_n(0)\|_H^2 \le -\frac{\alpha}{2}\int_0^t \|u_n(s)\|_V^2 \, ds \; + \; \frac{1}{2\alpha}\int_0^t \|f(s)\|_{V'}^2 \, ds.$$

Equivalently,

$$\|u_n(t)\|_H^2 \; + \; \alpha \int_0^t \|u_n(s)\|_V^2 \, ds \le \|u_n(0)\|_H^2 \; + \; \frac{1}{\alpha}\int_0^t \|f(s)\|_{V'}^2 \, ds.$$

Replacing t by the supremum over $[0, T]$ and weakening the constants, we obtain the a priori estimate

$$\sup_{t\in[0,T]} \|u_n(t)\|_H^2 \; + \; \frac{\alpha}{2}\int_0^T \|u_n(t)\|_V^2 \, dt \le \|u_n(0)\|_H^2 + \frac{1}{2\alpha}\int_0^T \|f(t)\|_{V'}^2 \, dt.$$

Since $u_n(0)$ is the H-orthogonal projection of u_0 onto V_n, $\|u_n(0)\|_H \le \|u_0\|_H$, and the right-hand side is independent of n. Using this uniform bound and weak compactness, we may extract a subsequence, again denoted by (u_n), such that

$$u_n \rightharpoonup u \quad \text{in } L^2([0, T]; V).$$

If $A \in \mathcal{L}(V, V')$ is linear, then also

$$Au_n \rightharpoonup Au \quad \text{in } L^2([0, T]; V').$$

Indeed, let $A^* : V \to V'$ denote the adjoint operator, characterized by

$${}_{V'}\langle Aw, v\rangle_V = {}_{V'}\langle A^* v, w\rangle_V \qquad \text{for all } w, v \in V.$$

Then for any $v \in L^2([0, T]; V)$,

$$\int_0^T {}_{V'}\langle Au_n(t), v(t)\rangle_V \, dt = \int_0^T {}_{V'}\langle A^* v(t), u_n(t)\rangle_V \, dt.$$

Since $A^* \in \mathcal{L}(V, V')$, we have $A^* v \in L^2([0, T]; V')$, and by weak convergence of u_n in $L^2([0, T]; V)$,

$$\int_0^T {}_{V'}\langle A^* v(t), u_n(t)\rangle_V \, dt \longrightarrow \int_0^T {}_{V'}\langle A^* v(t), u(t)\rangle_V \, dt \qquad \text{for all } v \in L^2([0, T]; V).$$

Applying the adjoint identity once more, this limit equals

$$\int_0^T {}_{V'}\langle Au(t), v(t)\rangle_V \, dt.$$

This allows one to pass to the limit in the variational formulation and obtain a solution of the abstract parabolic problem.

Generalization to nonlinear operators

Assume now that $A : V \to V'$ is still bounded and coercive but *not necessarily linear*. The same a priori estimate remains true, so after extracting a subsequence we may assume

$$u_n \rightharpoonup u \quad \text{in } L^2([0,T];V).$$

Moreover, the growth bound on A implies that $(A(u_n))$ is bounded in $L^2([0,T];V')$, so, after passing to a further subsequence if necessary,

$$A(u_n) \rightharpoonup g \quad \text{in } L^2([0,T];V')$$

for some (a priori unknown) g. In contrast to the linear case, weak convergence $u_n \rightharpoonup u$ does *not* directly imply $A(u_n) \rightharpoonup A(u)$, so we need an additional argument to identify g.

Monotonicity (dissipativity)

Assume the following one-sided Lipschitz/monotonicity condition: there exists $\lambda \in \mathbb{R}$ such that for all $u, v \in V$,

$${}_{V'}\langle A(u) - A(v), u - v\rangle_V \le \lambda \|u - v\|_H^2.$$

Together with this, assume the growth bound

$$\|A(u)\|_{V'} \le M(1 + \|u\|_V), \qquad u \in V,$$

and the hemicontinuity condition: for all $u, v, w \in V$, the map $\theta \mapsto {}_{V'}\langle A(u+\theta v), w\rangle_V$ is continuous on $\mathbb{R}$.

For any $v \in L^2([0,T];V)$, the one-sided Lipschitz/monotonicity condition gives, after integrating in time,

$$\int_0^T {}_{V'}\langle A(u_n(t)) - A(v(t)), u_n(t) - v(t)\rangle_V \, dt \le \lambda \int_0^T \|u_n(t) - v(t)\|_H^2 \, dt.$$

Expanding this inequality gives

$$\begin{aligned}
&\int_0^T {}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V \, dt \; - \int_0^T {}_{V'}\langle A(u_n(t)), v(t)\rangle_V \, dt \\
&\quad - \int_0^T {}_{V'}\langle A(v(t)), u_n(t)\rangle_V \, dt \; + \int_0^T {}_{V'}\langle A(v(t)), v(t)\rangle_V \, dt \\
&\le \lambda \int_0^T \|u_n(t) - v(t)\|_H^2 \, dt.
\end{aligned}$$

From the energy estimate, one obtains the lim sup bound

$$\limsup_{n\to\infty} \int_0^T {}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V \, dt \le \int_0^T {}_{V'}\langle g(t), u(t)\rangle_V \, dt.$$

Also,

$$\int_0^T {}_{V'}\langle A(u_n(t)), v(t)\rangle_V \, dt \longrightarrow \int_0^T {}_{V'}\langle g(t), v(t)\rangle_V \, dt$$

by weak convergence of $A(u_n)$ in $L^2([0,T];V')$, and

$$\int_0^T {}_{V'}\langle A(v(t)), u_n(t)\rangle_V \, dt \longrightarrow \int_0^T {}_{V'}\langle A(v(t)), u(t)\rangle_V \, dt$$

because $A(v) \in L^2([0,T];V')$ and $u_n \rightharpoonup u$ in $L^2([0,T];V)$. Passing to the limit yields

$$\int_0^T {}_{V'}\langle g(t) - A(v(t)), u(t) - v(t)\rangle_V \, dt \le \lambda \int_0^T \|u(t) - v(t)\|_H^2 \, dt \qquad \forall\, v \in L^2([0,T];V).$$

The remaining step is the classical *Browder–Minty (Minty) trick*: we perturb v around u and let the perturbation parameter tend to zero, using hemicontinuity to pass to the limit in the nonlinear term.

Now choose

$$v = u + \theta w, \qquad \theta > 0, \quad w \in L^2([0,T];V).$$

Then $u - v = -\theta w$ and therefore

$$-\theta \int_0^T {}_{V'}\langle g(t) - A(u(t) + \theta w(t)), w(t)\rangle_V \, dt \le \lambda\theta^2 \int_0^T \|w(t)\|_H^2 \, dt.$$

Dividing by $\theta > 0$ and multiplying by -1 yields

$$\int_0^T {}_{V'}\langle g(t) - A(u(t) + \theta w(t)), w(t)\rangle_V \, dt \ge -\lambda\theta \int_0^T \|w(t)\|_H^2 \, dt.$$

Letting $\theta \downarrow 0$ and using hemicontinuity gives

$$\int_0^T {}_{V'}\langle g(t) - A(u(t)), w(t)\rangle_V \, dt \ge 0 \qquad \forall\, w \in L^2([0,T];V).$$

Replacing w by $-w$ gives

$$\int_0^T {}_{V'}\langle g(t) - A(u(t)), -w(t)\rangle_V \, dt \ge 0, \qquad \forall\, w \in L^2([0,T];V),$$

equivalently,

$$\int_0^T {}_{V'}\langle g(t) - A(u(t)), w(t)\rangle_V \, dt \le 0, \qquad \forall\, w \in L^2([0,T];V).$$

Therefore,

$$\int_0^T {}_{V'}\langle g(t) - A(u(t)), w(t)\rangle_V \, dt = 0 \qquad \forall\, w \in L^2([0,T];V),$$

hence $g = A(u)$ in $L^2([0,T];V')$.

We now relate the quantity $\int_0^T {}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V \, dt$ to the energy balance and use it to compare $\int_0^T {}_{V'}\langle g(t), u(t)\rangle_V \, dt$ with $\liminf_{n\to\infty} \int_0^T {}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V \, dt$.

First, the Galerkin equation can be written as $u_n'(t) = A(u_n(t)) + f(t)$ in V_n. Testing it by $u_n(t)$ and using $\frac{1}{2}\frac{d}{dt}\|u_n(t)\|_H^2 = {}_{V'}\langle u_n'(t), u_n(t)\rangle_V$ gives

$${}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V = \frac{1}{2}\frac{d}{dt}\|u_n(t)\|_H^2 \; - \; {}_{V'}\langle f(t), u_n(t)\rangle_V.$$

Integrating over $[0,T]$ yields

$$\int_0^T {}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V \, dt = \frac{1}{2}\big(\|u_n(T)\|_H^2 - \|u_n(0)\|_H^2\big) \; - \; \int_0^T {}_{V'}\langle f(t), u_n(t)\rangle_V \, dt.$$

Next, passing to the limit in the Galerkin formulation yields (in V')

$$u'(t) = g(t) + f(t),$$

meaning that for a.e. $t \in [0,T]$ and all $\varphi \in V$,

$$_{V'}\langle u'(t), \varphi\rangle_V = {}_{V'}\langle g(t), \varphi\rangle_V + {}_{V'}\langle f(t), \varphi\rangle_V.$$

Choosing $\varphi = u(t)$, integrating over $[0,T]$, and using $\frac{1}{2}\frac{d}{dt}\|u(t)\|_H^2 = {}_{V'}\langle u'(t), u(t)\rangle_V$, we obtain

$$\int_0^T {}_{V'}\langle g(t), u(t)\rangle_V \, dt = \frac{1}{2}\big(\|u(T)\|_H^2 - \|u(0)\|_H^2\big) \; - \; \int_0^T {}_{V'}\langle f(t), u(t)\rangle_V \, dt.$$

Finally, extract a subsequence such that $u_n(T) \rightharpoonup u(T)$ in H. Since $u_n \rightharpoonup u$ in $L^2([0,T];V)$, we have

$$\int_0^T {}_{V'}\langle f(t), u_n(t)\rangle_V \, dt \longrightarrow \int_0^T {}_{V'}\langle f(t), u(t)\rangle_V \, dt,$$

and since $u_n(0) \to u(0)$ in H (Galerkin projection of the common initial datum) together with weak lower semicontinuity of the norm in H, we infer

$$\begin{aligned}
\int_0^T {}_{V'}\langle g(t), u(t)\rangle_V \, dt &\le \liminf_{n\to\infty} \left[\frac{1}{2}\big(\|u_n(T)\|_H^2 - \|u_n(0)\|_H^2\big) \; - \; \int_0^T {}_{V'}\langle f(t), u_n(t)\rangle_V \, dt\right] \\
&= \liminf_{n\to\infty} \int_0^T {}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V \, dt \\
&\le \limsup_{n\to\infty} \int_0^T {}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V \, dt.
\end{aligned}$$

Thus we have linked the limit quantity $\int_0^T {}_{V'}\langle g(t), u(t)\rangle_V \, dt$ to the lim inf / lim sup of $\int_0^T {}_{V'}\langle A(u_n(t)), u_n(t)\rangle_V \, dt$. Combining this with the monotonicity inequality and passing to the limit in n yields, for every $v \in L^2([0,T];V)$,

$$\int_0^T {}_{V'}\langle g(t) - A(u(t)), u(t) - v(t)\rangle_V \, dt \le \limsup_{n\to\infty} \int_0^T {}_{V'}\langle A(u_n(t)) - A(v(t)), u(t) - v(t)\rangle_V \, dt \le 0.$$

This variational inequality is the key output of the Browder–Minty/Minty trick and is the step that leads to the identification $g = A(u)$.

Compactness

The monotonicity method above is not the only way to identify nonlinear terms in a limit. Another important approach is to use *compactness* to upgrade weak convergence to strong convergence in an intermediate space.

Suppose that (u_n) is bounded in

$$L^2([0,T];V).$$

In practice one also needs a bound that prevents fast oscillations in time. A common formulation is a fractional time-regularity estimate such as

$$u_n \text{ bounded in } L^2([0,T];V) \cap H^{1/2}([0,T];V'),$$

or, in many PDE/SPDE settings, a bound on the time derivative u_n' in a negative space (e.g. u_n' bounded in $L^2([0,T];V')$). If the embedding $V \hookrightarrow H$ is compact, then (under additional time-regularity bounds, e.g. on (u_n') in a negative space) one can often extract a subsequence that converges *strongly* in

$$L^2([0,T];H).$$

Statements of this type are instances of the Aubin–Lions/Simon compactness lemma: compactness in space plus some regularity in time yields compactness in space–time. Recall that the embedding $V \hookrightarrow H$ is called *compact* if the inclusion map $i : V \to H$, $i(v) = v$, is a compact linear operator, i.e. it maps bounded sets in V to *relatively compact* sets in H. Here "relatively compact in H" means that every sequence in the set admits a subsequence that converges in H (equivalently, the closure of the set is compact in H).

Remark 9 (PDE example: Rellich–Kondrachov). For a bounded domain $D \subset \mathbb{R}^d$,

$$H_0^1(D) \hookrightarrow L^2(D)$$

is a compact embedding (Rellich–Kondrachov theorem): any sequence bounded in $H_0^1(D)$ admits a subsequence converging strongly in $L^2(D)$. This is a prototypical source of compactness in variational PDE/SPDE theory.

Remark 10 (Why compactness helps). Strong convergence in $L^2([0,T];H)$ is especially useful when the equation contains nonlinearities that are not monotone (e.g. convection terms in fluid dynamics such as Navier–Stokes). In such cases, weak convergence may be insufficient to pass to the limit in products, while strong convergence can restore convergence of nonlinear terms.